HDOC Day-43

String Operations Using a Menu-Driven C Program

Aim

To write a menu-driven C program using separate user-defined functions to find the longest word in a sentence, determine whether one string is a rotation of another, and exit.

Algorithm

1. Start the program and display a menu with three options: longest word, string rotation, and exit.

2. Accept a menu choice and call the appropriate function using switch-case.

3. For longest word, read a sentence with fgets(), remove its newline, and copy it into another array.

4. Separate words using strtok(), compare their lengths with strlen(), and store the longest using strcpy(). Display the word and its length.

5. For rotation, read two strings, remove their trailing newlines, and compare their lengths.

6. If the lengths match, concatenate the first string with itself and search for the second using strstr(). Display whether the strings are rotations.

7. Display an error for an invalid choice and repeat until Exit is selected.

C Program

```
#include <stdio.h>
#include <string.h>

#define MAX 200

void findLongestWord(void);
void checkRotation(void);

int main()
{
    int choice;

    do
    {
        printf("\n----- STRING OPERATIONS MENU -----\n");
        printf("1. Find Longest Word in a Sentence\n");
        printf("2. Check String Rotation\n");
        printf("3. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                findLongestWord();
                break;
            case 2:
                checkRotation();
                break;
            case 3:
                printf("Exiting program...\n");
                break;
            default:
                printf("Invalid choice! Please enter 1 to 3.\n");
        }
    } while (choice != 3);

    return 0;
}

/* Function to find longest word */
void findLongestWord(void)
{
    char sentence[MAX];
    char copy[MAX];
    char longest[MAX] = "";
    char *word;

    printf("Enter a sentence: ");
    fgets(sentence, MAX, stdin);
    sentence[strcspn(sentence, "\n")] = '\0';

    strcpy(copy, sentence);
    word = strtok(copy, " \t");

    while (word != NULL)
    {
        if (strlen(word) > strlen(longest))
            strcpy(longest, word);
        word = strtok(NULL, " \t");
    }
```

```
    if (strlen(longest) == 0)
        printf("No words found.\n");
    else
    {
        printf("Longest Word: %s\n", longest);
        printf("Length: %zu\n", strlen(longest));
    }
}

/* Function to check string rotation */
void checkRotation(void)
{
    char str1[MAX], str2[MAX];
    char combined[2 * MAX];

    printf("Enter first string: ");
    fgets(str1, MAX, stdin);
    printf("Enter second string: ");
    fgets(str2, MAX, stdin);

    str1[strcspn(str1, "\n")] = '\0';
    str2[strcspn(str2, "\n")] = '\0';

    if (strlen(str1) != strlen(str2))
    {
        printf("The strings are not rotations.\n");
        return;
    }

    strcpy(combined, str1);
    strcat(combined, str1);

    if (strstr(combined, str2) != NULL)
        printf("The strings are rotations of each other.\n");
    else
        printf("The strings are not rotations.\n");
}
```

Test Case Table

Test	Operation	Input	Expected result
1	Longest word	C programming is interesting	programming (11)
2	Longest word	I love computers	computers (9)
3	Longest word	Hello world	Hello (5)
4	Longest word	Empty sentence	No words found
5	Rotation	abcd / cdab	Rotations
6	Rotation	hello / llohe	Rotations
7	Rotation	abcd / acbd	Not rotations
8	Rotation	abc / abcd	Not rotations
9	Invalid menu	4	Invalid choice message
10	Exit	3	Program exits

Sample Output

```
----- STRING OPERATIONS MENU -----
1. Find Longest Word in a Sentence
2. Check String Rotation
3. Exit
Enter your choice: 1
Enter a sentence: C programming is interesting
Longest Word: programming
Length: 11

Enter your choice: 2
Enter first string: abcd
Enter second string: cdab
The strings are rotations of each other.

Enter your choice: 2
Enter first string: hello
Enter second string: world
The strings are not rotations.

Enter your choice: 4
Invalid choice! Please enter 1 to 3.

Enter your choice: 3
Exiting program...
```

Result

The menu-driven C program uses separate user-defined functions, character arrays, loops, conditional statements, and a switch-case statement. It identifies the longest word in a sentence and checks whether one string is a rotation of another. The menu repeats until Exit is selected.